UNISOC

Unisoc Confidential For longcheer

UMS9230

Audio Test Report

WWW.UNISOC.COM

UNISOC (SHANGHAI) TECHNOLOGIES CO., LTD.

Keyword

Keyword: Audio, Test Report.

Unisoc Confidential For longcheer

Unisoc Confidential For longcheer

Contents

01 Test Outline

02 Test Result

Unisoc Confidential For longcheer

01

Test Outline

Test Outline

Unisoc Confidential For longcheer

Project Name： UMS9230

HW Version： UMS9230_1_IRD__A_V1.0.0

Content Outline：

Test case		Remark
14002 Multimedia Test	14002001 Handsfree Music	PASS
	14002002 Headset Class G Music	PASS
	14002017 Pop Noise	PASS
14003 Voice Call	14003001 Handset Voice Call NB	PASS
	14003002 Handfree Voice Call NB	PASS
	14003018 Handset VOLTE_WB	PASS
	14003019 Handfree_VOLTE_WB	PASS

Unisoc Confidential For longcheer

02

Test Result

Test Result

UNISOC

14002001 Handsfree Music

(Ext PA: UCP1301, AGC on)

Unisoc Confidential For longcheer

Test Item	Spec	Result
Output level @Max level	≥1.8V(RMS)	2.74V
SNR@Max level	≤-83dB	-96.4dB
THD+N@1K Hz	≤-40dB(1%)	-73.7dB
Noise@ Level 1	≤45uV	40.1uV
Dynamic Range	NA	-93.6dB

Test Result

UNISOC

14002001 Handsfree Music

(Ext PA: UCP1301, AGC off)

Unisoc Confidential For longcheer

Test Item	Spec	Result
Output level @Max level	≥1.8V(Rms)	5.1V
SNR@Max level	≤-83dB	-102.9dB
THD+N@1K Hz	≤-40dB(1%)	-70.8dB
Noise@ Level 1	≤45uV	37.6uV
Dynamic Range	NA	-93.5dB

Test Result

UNISOC

14002002 Headset Class G Music

Unisoc Confidential For longcheer

Test Item	Spec	Result
Output level@Max_L	400±50mV(Rms)	403.5mV
Output level@Max_R	400±50mV(Rms)	403.6mV
SNR@Max Volume	≤-83dB	-105.3dB
SNR@30mV	≤-68dB	-84.3dB
THD+N@1KHz_L	≤-70dB	-86.3dB
THD+N@1KHz_R	≤-70dB	-87.6dB
Noise@Level 1	≤12uV	2.44uV
Cross Talk@1Khz(With FM)	≤-50dB	-50.1dB

Test Result

UNISOC

14002017 Pop Noise

Unisoc Confidential For longcheer

Test Item	Spec	Result
Headset turn on Music player	≤ 0.3mV	0.07mV
Headset turn off Music player	≤ 0.3mV	0.07mV
Handsfree turn on Music player	≤ 3mV	0.65mV
Handsfree turn off Music player	≤ 3mV	0.5mV
Plug in headphone	≤ 3mV	0.01mV
Plug out headphone	≤ 3mV	1.3mV

Test Result

UNISOC

14003001 Handset Voice Call GSM_NB

Unisoc Confidential For longcheer

Test Item	Spec	Result
RLR@Max Vol	-3~-10dB	-7.5dB
RLR@nom Vol	2 ± 3dB	1.3dB
SLR	8 ± 3dB	7.8dB
RFR@Nom Vol	In mark	Pass
SFR	In mark	Pass
Side tone	≥ 13dB	23.4dB
Receiving distortion @Max vol	In mask	Pass
Sending distortion	In mask	Pass
Receiving Idle channel noise@Max Vol	≤ -54 dBPa(A)	-64.6dBPa(A)
Receiving Idle channel noise@Nom Vol	≤ -57 dBPa(A)	-65.2dBPa(A)
Sending Idle channel noise	≤ -64 dBm0(P)	-87.7dBm0(P)
Receiving TMOS @Max Vol	≥3.0	3.72
Sending TMOS	≥3.0	4.03
TCL(Echo Loss)@Max Vol	≥55dB	66.5dB
TCL(Echo Loss)@Nom Vol	≥55dB	65.1dB
3Quest_S-MOS@Dual Mic	≥3.8	3.93
3Quest_N-MOS@Dual Mic	≥3.8	3.99

Test Result

UNISOC

14003002 Handfree Voice Call GSM_NB

Unisoc Confidential For longcheer

Test Item	Spec	Result
RLR@Max Vol	2~16dB	9.0dB
SLR	13±4dB	13.4dB
Sending distortion	In mask	Pass
Receiving Idle channel noise@Max Vol	≤ -54 dBPa(A)	-69.3dBPa(A)
Receiving Idle channel noise@Nom Vol	≤ -57 dBPa(A)	-69.8dBPa(A)
Sending Idle channel noise	≤ -64 dBm0(P)	-69.3dBm0(P)
Receiving TMOS @Nom Vol	≥1.8	3.20
Sending TMOS	≥3	3.95
TCL(Echo Loss)@Max Vol	≥46dB	71.7dB

Test Result

UNISOC

14003018 Handset VOLTE_WB (Rate：23.85kbps)

Unisoc Confidential For longcheer

Test Item	Spec	Result
RLR@Max Vol	-3~-10dB	-6.6dB
RLR@nom Vol	2 ± 3dB	2.1dB
SLR	8 ± 3dB	8.5dB
RFR@Nom Vol	In mark	Pass
SFR	In mark	Pass
Side tone	≥ 13dB	16.4dB
Receiving distortion @Max vol	In mask	Pass
Sending distortion	In mask	Pass
Receiving Idle channel noise@Max Vol	≤ -54 dBPa(A)	-67.7dBPa(A)
Receiving Idle channel noise@Nom Vol	≤ -57 dBPa(A)	-68.5dBPa(A)
Sending Idle channel noise	≤ -64 dBm0(P)	-77.9dBm0(P)
Receiving TMOS @Nom Vol	≥3.0	3.41
Sending TMOS	≥3.0	4.22
TCL(Echo Loss)@Max Vol	≥55dB	63.6dB
TCL(Echo Loss)@Nom Vol	≥55dB	64.3dB
3Quest_S-MOS@Dual Mic	≥3.8	4.07
3Quest_N-MOS@Dual Mic	≥3.8	3.96

Test Result

UNISOC

14003019 Handfree_VOLTE_WB (Rate：23.85kbps)

Unisoc Confidential For longcheer

Test Item	Spec	Result
RLR@Max Vol	2~16dB	10.0dB
SLR	13±4dB	13.2dB
Sending distortion	In mask	Pass
Receiving Idle channel noise@Max Vol	≤ -54 dBPa(A)	-69.2dBPa(A)
Sending Idle channel noise	≤ -64 dBm0(P)	-70.5dBm0(P)
Receiving TMOS @Nom Vol	≥1.8	2.38
Sending TMOS	≥3	3.89
TCL(Echo Loss)@Max Vol	≥46dB	64.3dB

Unisoc Confidential For longcheer

Thank You

UNISOC®

All data and information contained in or disclosed by this document is confidential and proprietary information of UNISOC (Shanghai) Technologies Co., Ltd. (hereafter referred as UNISOC) and all rights therein are expressly reserved. This document is provided for reference purpose, no license (express or implied, by estoppel or otherwise) to any intellectual property rights is granted by this document, and no express and implied warranties, including but without limitation, the implied warranties of fitness for any particular purpose, and non-infringement, as well as any performance. By accepting this material, the recipient agrees that the material and the information contained therein is to be held in confidence and in trust and will not be used, copied, reproduced in whole or in part, nor its contents revealed in any manner to others without the express written permission of UNISOC. UNISOC may make any changes at any time without prior notice. Although every reasonable effort is made to present current and accurate information, UNISOC makes no guarantees of any kind with respect to the matters addressed in this document. In no event shall UNISOC be responsible or liable, directly or indirectly, for any damage or loss caused or alleged to be caused by or in connection with the use of or reliance on any such content.

Please refer to the UNISOC Documents in the UNISOC Deliverables for the use of the Deliverables. Any loss caused by the modification, customization or use of the UNISOC Deliverables in violation of the instructions in the UNISOC Documents shall be undertaken by those who conduct so. The performance indicators, test results and parameters in the UNISOC Deliverables are all obtained in the internal development and test system of UNISOC and are only for the reference. Before using UNISOC Deliverables commercially or conducting mass production of the Deliverables, comprehensive testing and debugging in combination with its own software and hardware test environment are pre-requisite.